

Teorema de la convergencia dominada

Teorema 1 (de la convergencia dominada). Sea $(f_n)_{n \in \mathbb{N}}$ una sucesión de fns medibles en (X, Σ, μ) tal que $\exists \lim f_n =: f \wedge \exists g$ integrable : $\forall n \in \mathbb{N} : |f_n| \leq g^1$

$$\implies \lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0, \quad \text{en particular,} \quad \int \lim_{n \rightarrow \infty} f_n d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

Demostración: Consideramos $(\hat{f}_n)_{n \in \mathbb{N}}$ dada por $\forall n \in \mathbb{N} : \hat{f}_n = f_n - f \implies \hat{f}_n \xrightarrow{n \rightarrow \infty} 0$.

Definimos $\forall n \in \mathbb{N} : h_n := g - |\hat{f}_n| \geq 0 \xRightarrow{\text{Fatou}} \liminf_{n \rightarrow \infty} h_n d\mu \leq \liminf_{n \rightarrow \infty} \int h_n d\mu$.

Ahora, como $\liminf_{n \rightarrow \infty} h_n = \lim_{n \rightarrow \infty} g - |\hat{f}_n| = g$ porque $\hat{f}_n \xrightarrow{n \rightarrow \infty} 0$, entonces

$$\begin{aligned} \int g d\mu &\leq \liminf_{n \rightarrow \infty} \int (g - |\hat{f}_n|) d\mu \stackrel{\text{linealidad}}{=} \liminf_{n \rightarrow \infty} \left(\int g d\mu - \int |\hat{f}_n| d\mu \right) \\ &\implies \int g d\mu \leq \liminf_{n \rightarrow \infty} \int g d\mu - \limsup_{n \rightarrow \infty} \int |\hat{f}_n| d\mu \end{aligned}$$

Como $\liminf_{n \rightarrow \infty} \int g d\mu = \int g d\mu \wedge g \in \mathcal{L}^1(\mu)$, podemos cancelar y nos queda

$$\cancel{\int g d\mu} \leq \cancel{\int g d\mu} - \limsup_{n \rightarrow \infty} \int |\hat{f}_n| d\mu \implies \limsup_{n \rightarrow \infty} \int |\hat{f}_n| d\mu \leq 0$$

Lo que implica que $\limsup_{n \rightarrow \infty} \int |f_n - f| d\mu = 0 \implies \lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.

$$\text{y como } \int |f_n - f| d\mu = \begin{cases} \int (f_n - f) d\mu = \int f_n d\mu - \int f d\mu & \text{si } f_n \geq f \\ \int (f - f_n) d\mu = \int f d\mu - \int f_n d\mu & \text{si } f_n < f \end{cases}$$

se tiene $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0 \iff \lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu = \int \lim_{n \rightarrow \infty} f_n d\mu$. ■

¹Se dice que g “domina” a todas las f_n .

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